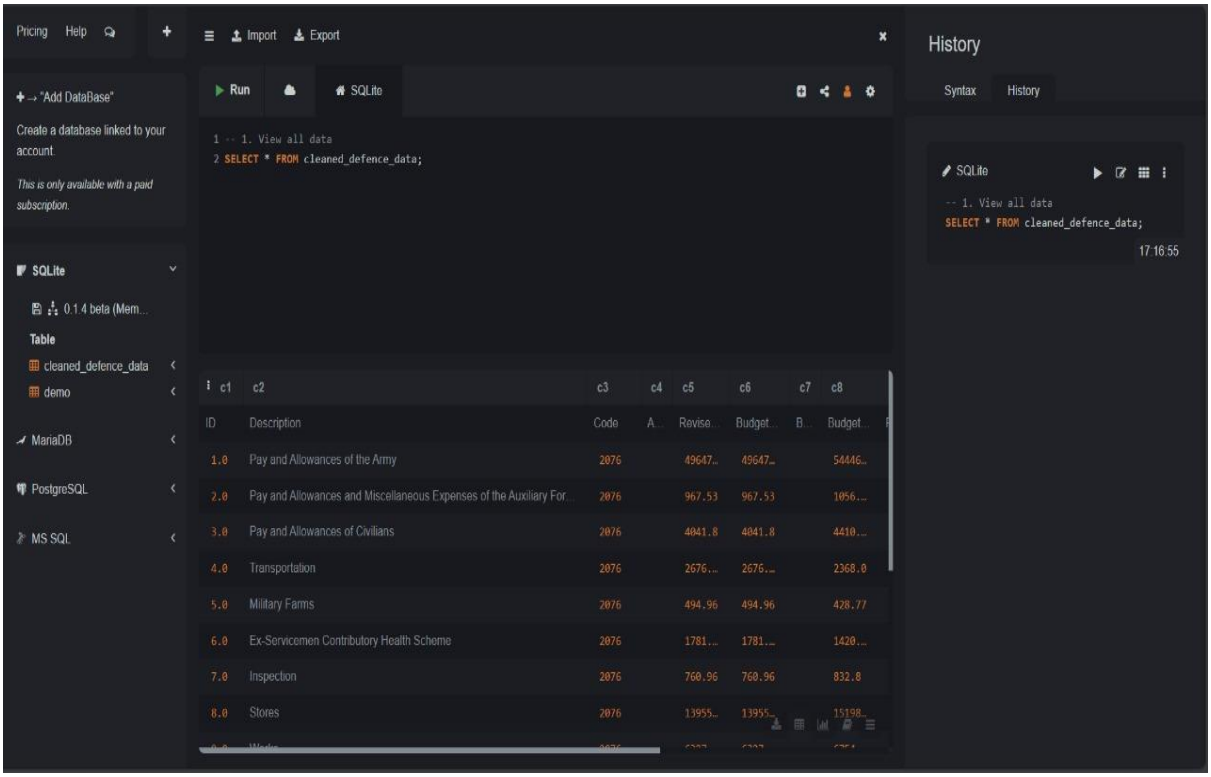
# Task-3: SQL For Data Analysis

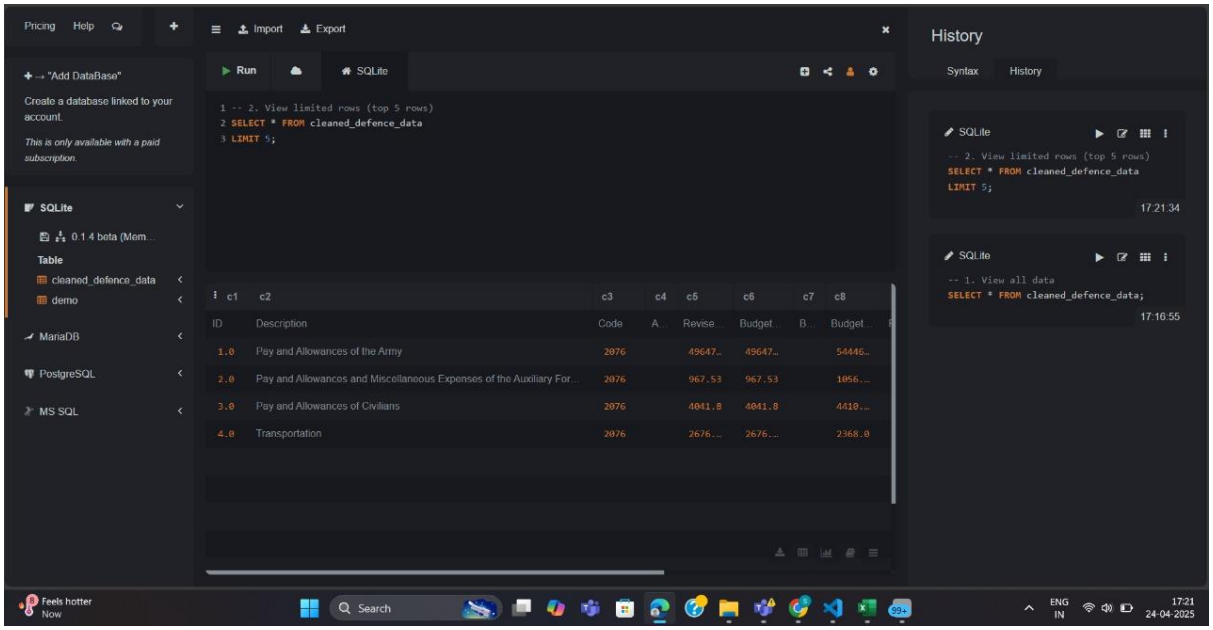
Objective: Use SQL queries to extract and analyze data from a database.

Tools: MySQL or PostgreSQL or SQLite

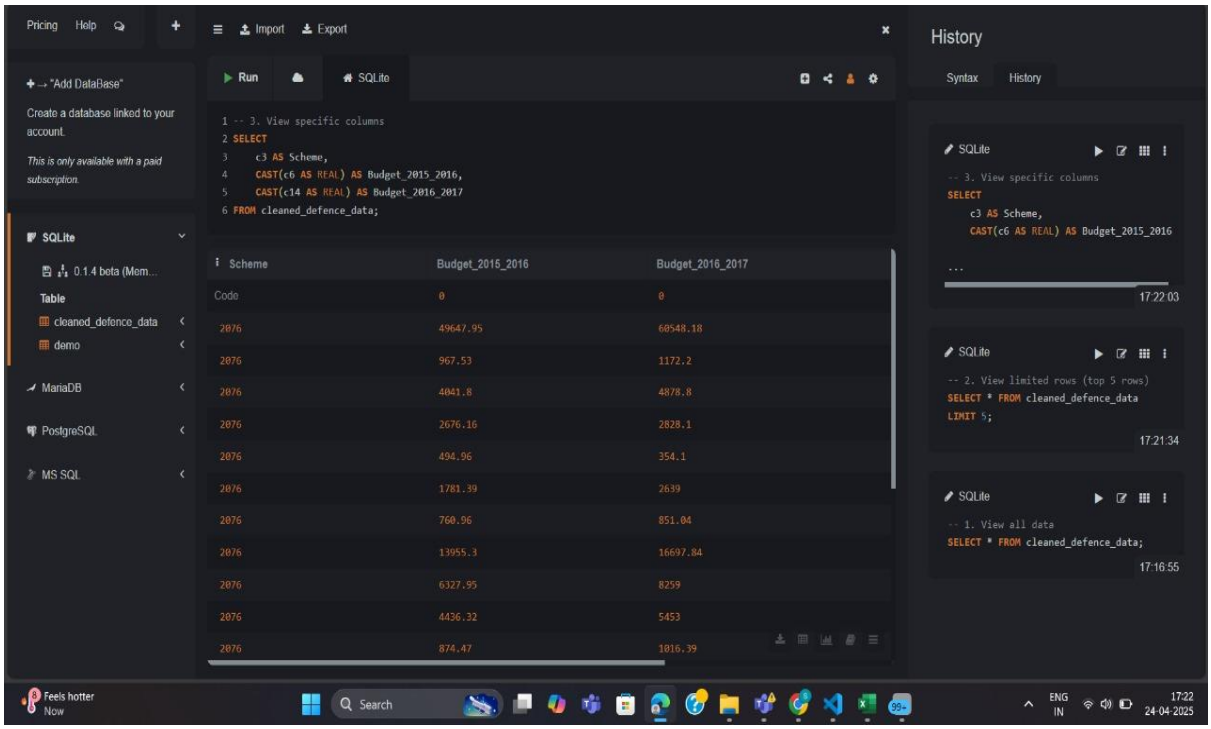
Deliverables: SQL queries in a SQL file + screenshots of output



**2.**


Pricing
Help
Import
Export
Run
SQLite
"Add DataBase"
Create a database linked to your account.
This is only available with a paid subscription.
SQLite
0.1.4 beta (Mem...
Table
cleaned_defence_data
demo
MariaDB
PostgreSQL
MS SQL
1 -- 2. View limited rows (top 5 rows)
2 SELECT * FROM cleaned_defence_data
3 LIMIT 5;
c1 c2 c3 c4 c5 c6 c7 c8
ID Description Code A... Revise... Budget... B... Budget...
1.0 Pay and Allowances of the Army 2076 49647... 49647... 54446...
2.0 Pay and Allowances and Miscellaneous Expenses of the Auxiliary For... 2076 967.53 967.53 1056...
3.0 Pay and Allowances of Civilians 2076 4041.8 4041.8 4410...
4.0 Transportation 2076 2676... 2676... 2368.0
History
Syntax
History
SQLite
-- 2. View limited rows (top 5 rows)
SELECT * FROM cleaned_defence_data
LIMIT 5;
17:21:34
SQLite
-- 1. View all data
SELECT * FROM cleaned_defence_data;
17:16:55
Feels hotter Now
Search
ENG IN
17:21 24-04-2025

**3.**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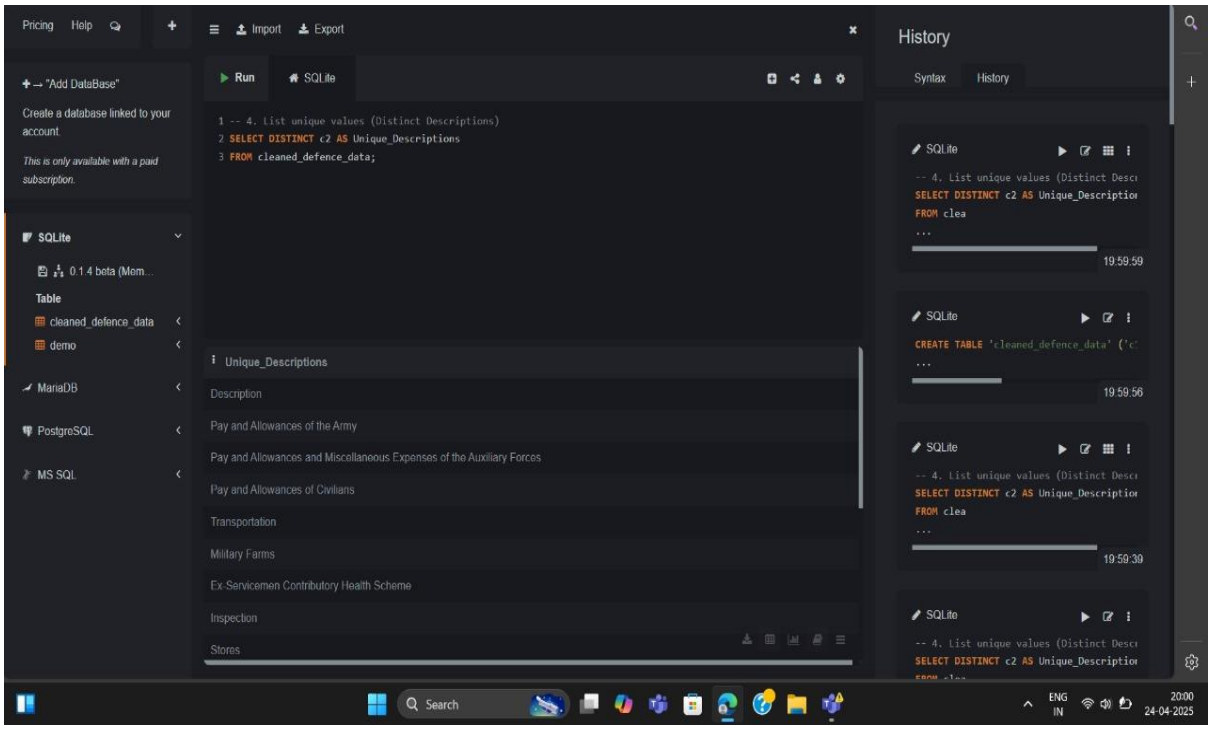
SQLite
0.1.4 beta (Mem...
Table
cleaned_defence_data
demo
MariaDB
PostgreSQL
MS SQL
1 -- 3. View specific columns
2 SELECT
3 c3 AS Scheme,
4 CAST(c6 AS REAL) AS Budget_2015_2016,
5 CAST(c14 AS REAL) AS Budget_2016_2017
6 FROM cleaned_defence_data;
Scheme Budget_2015_2016 Budget_2016_2017
Code 0 0
2076 49647.95 60548.18
2076 967.53 1172.2
2076 4041.8 4878.8
2076 2676.16 2828.1
2076 494.96 354.1
2076 1781.39 2639
2076 760.96 851.04
2076 13955.3 16697.84
2076 6327.95 8259
2076 4436.32 5453
2076 874.47 1016.39
History
Syntax
History
SQLite
-- 3. View specific columns
SELECT
c3 AS Scheme,
CAST(c6 AS REAL) AS Budget_2015_2016
...
17:22:03
SQLite
-- 2. View limited rows (top 5 rows)
SELECT * FROM cleaned_defence_data
LIMIT 5;
17:21:34
SQLite
-- 1. View all data
SELECT * FROM cleaned_defence_data;
17:16:55
Feels hotter Now
Search
ENG IN
17:22 24-04-2025

4.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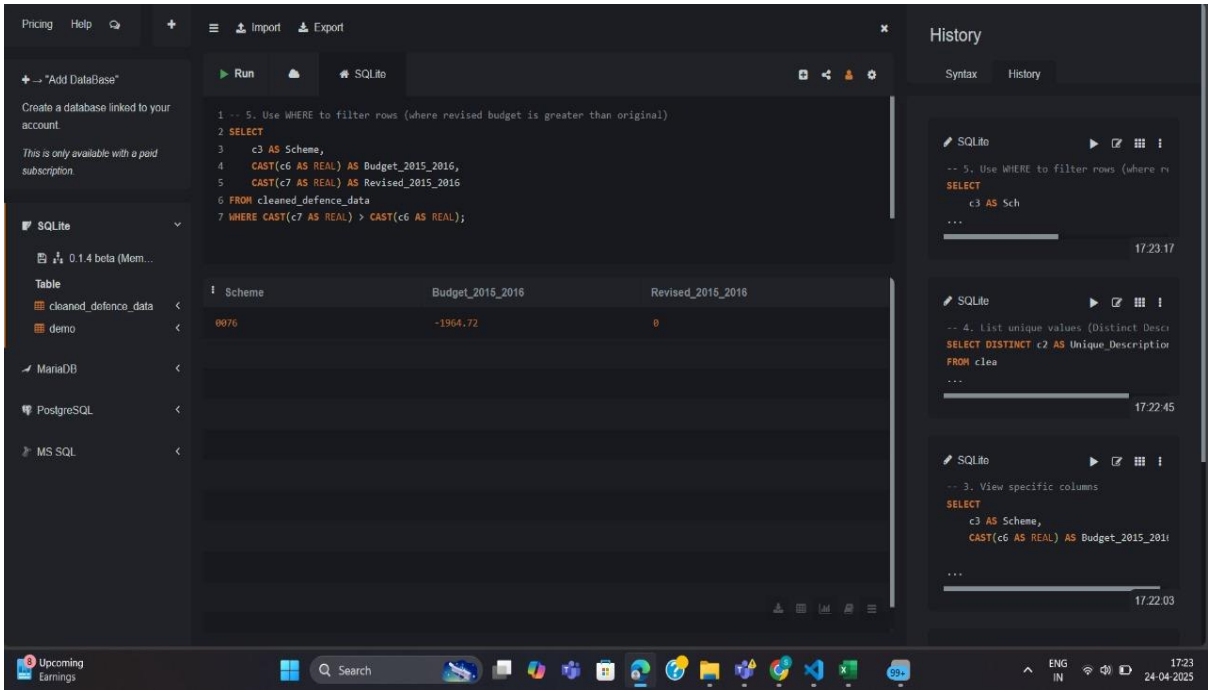
0.1.4 beta (Mem...
Table
cleaned_defence_data
demo
MariaDB
PostgreSQL
MS SQL
Run
SQLite
1 -- 4. List unique values (Distinct Descriptions)
2 SELECT DISTINCT c2 AS Unique_Descriptions
3 FROM cleaned_defence_data;
Unique_Descriptions
Description
Pay and Allowances of the Army
Pay and Allowances and Miscellaneous Expenses of the Auxiliary Forces
Pay and Allowances of Civilians
Transportation
Military Farms
Ex-Servicemen Contributory Health Scheme
Inspection
Stores
History
Syntax
History
19:59:59
19:59:56
19:59:30
Search
ENG
IN
2000
24-04-2025

5.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SQLite
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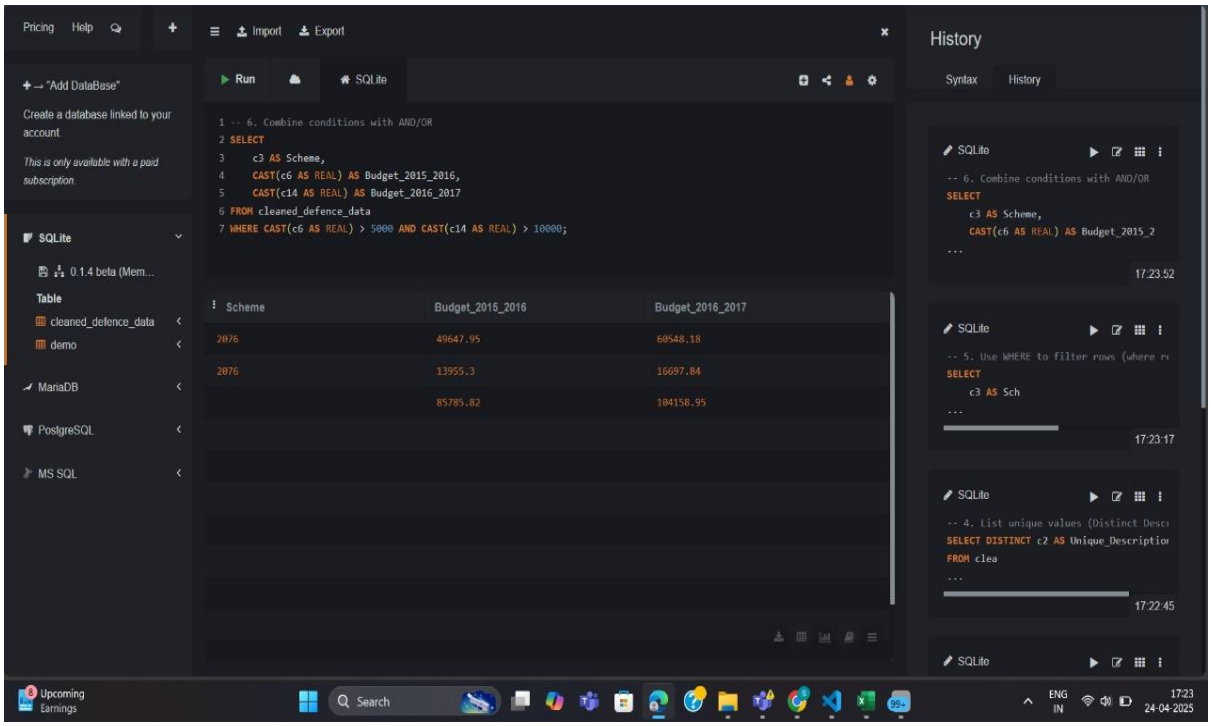
Table
cleaned_defence_data
demo
MariaDB
PostgreSQL
MS SQL
Run
SQLite
1 -- 5. Use WHERE to filter rows (where revised budget is greater than original)
2 SELECT
3 c3 AS Scheme,
4 CAST(c6 AS REAL) AS Budget_2015_2016,
5 CAST(c7 AS REAL) AS Revised_2015_2016
6 FROM cleaned_defence_data
7 WHERE CAST(c7 AS REAL) > CAST(c6 AS REAL);
Scheme
Budget_2015_2016
Revised_2015_2016
0076
-1964.72
0
History
Syntax
History
17:23:17
17:22:45
17:22:03
Upcoming Earnings
Search
ENG
IN
17:23
24-04-2025

6.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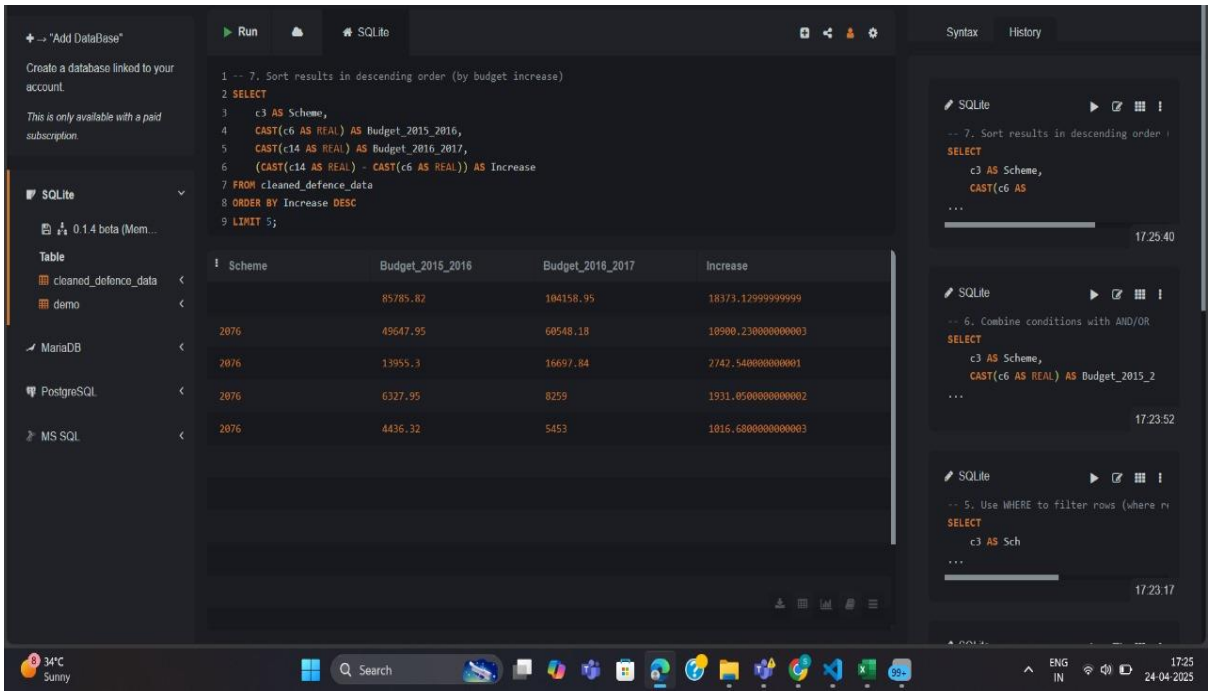
SQLite
0.1.4 beta (Mem...
Table
cleaned_defence_data
demo
MariaDB
PostgreSQL
MS SQL
Run
SQLite
1 -- 6. Combine conditions with AND/OR
2 SELECT
3 c3 AS Scheme,
4 CAST(c6 AS REAL) AS Budget_2015_2016,
5 CAST(c14 AS REAL) AS Budget_2016_2017
6 FROM cleaned_defence_data
7 WHERE CAST(c6 AS REAL) > 5000 AND CAST(c14 AS REAL) > 10000;
Scheme
Budget_2015_2016
Budget_2016_2017
2076
49647.95
60548.18
2076
13955.3
16697.84
85785.82
104158.95
History
Syntax
History
17:23:52
17:23:17
17:22:45
Upcoming Earnings
Search
ENG IN
17:23
24-04-2025

7.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Create a database linked to your account.
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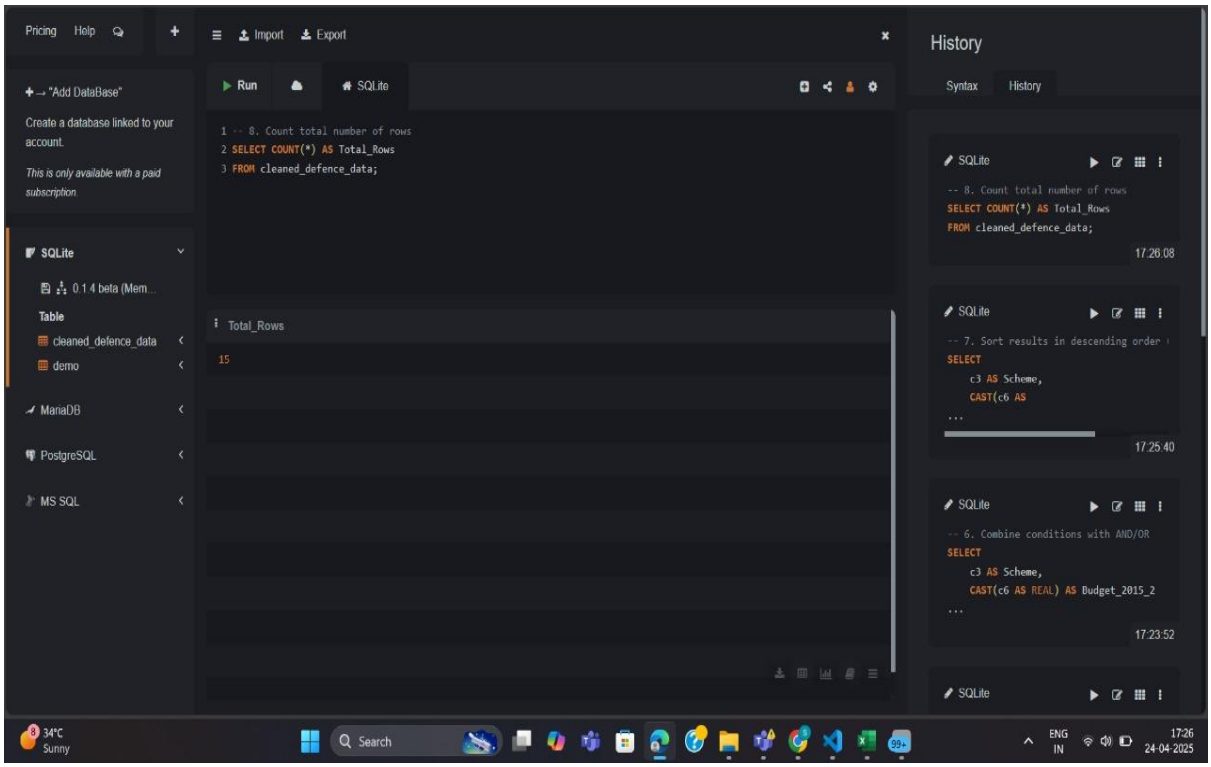
SQLite
0.1.4 beta (Mem...
Table
cleaned_defence_data
demo
MariaDB
PostgreSQL
MS SQL
Run
SQLite
1 -- 7. Sort results in descending order (by budget increase)
2 SELECT
3 c3 AS Scheme,
4 CAST(c6 AS REAL) AS Budget_2015_2016,
5 CAST(c14 AS REAL) AS Budget_2016_2017,
6 (CAST(c14 AS REAL) - CAST(c6 AS REAL)) AS Increase
7 FROM cleaned_defence_data
8 ORDER BY Increase DESC
9 LIMIT 5;
Scheme
Budget_2015_2016
Budget_2016_2017
Increase
85785.82
104158.95
18373.12999999999
2076
49647.95
60548.18
10900.230000000003
2076
13955.3
16697.84
2742.5400000000001
2076
6327.95
8259
1931.0500000000002
2076
4436.32
5453
1016.6800000000003
Syntax
History
17:25:40
17:23:52
17:23:17
34°C Sunny
Search
ENG IN
17:25
24-04-2025

8.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Import
Export
"Add DataBase"
Create a database linked to your account.
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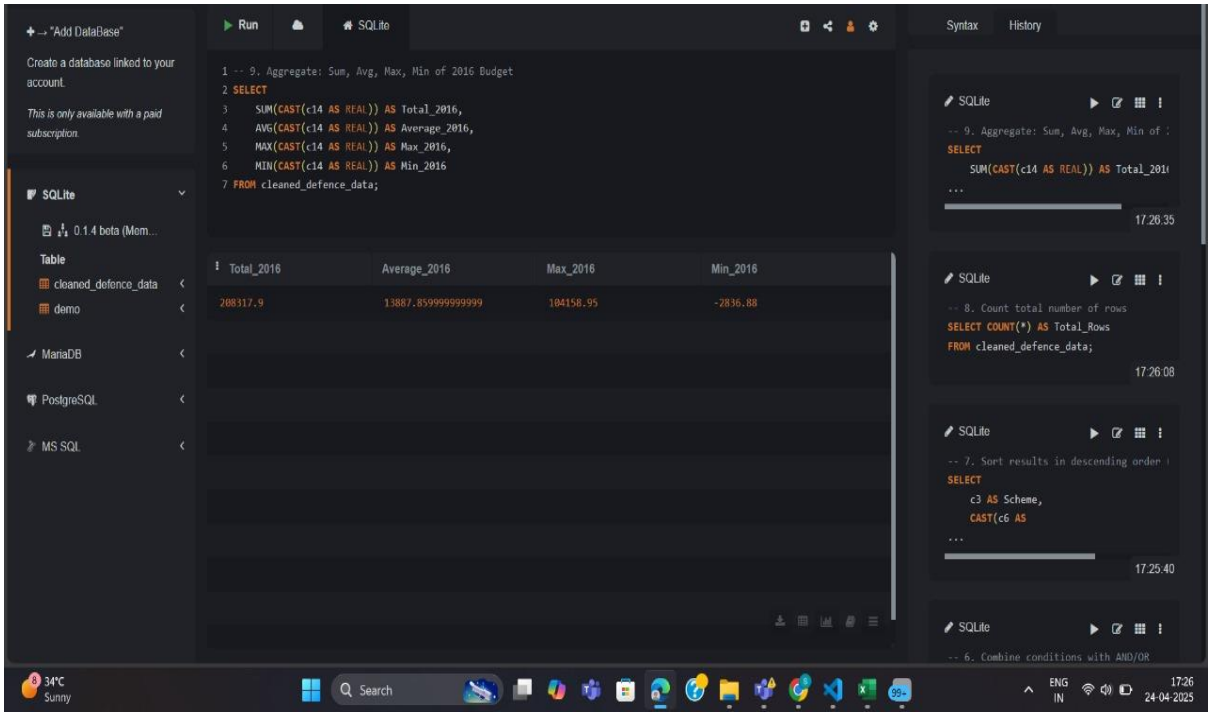
SQLite
0.1.4 beta (Mem...
Table
cleaned_defence_data
demo
MariaDB
PostgreSQL
MS SQL
Run
SQLite
1 -- 8. Count total number of rows
2 SELECT COUNT(*) AS Total_Rows
3 FROM cleaned_defence_data;
Total_Rows
15
History
Syntax
History
SQLite
-- 8. Count total number of rows
SELECT COUNT(*) AS Total_Rows
FROM cleaned_defence_data;
17:26:08
SQLite
-- 7. Sort results in descending order
SELECT
c3 AS Scheme,
CAST(c6 AS
...
17:25:40
SQLite
-- 6. Combine conditions with AND/OR
SELECT
c3 AS Scheme,
CAST(c6 AS REAL) AS Budget_2015_2
...
17:23:52
SQLite
34°C
Sunny
Search
ENG
IN
17:26
24-04-2025

9.


"Add DataBase"
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SQLite
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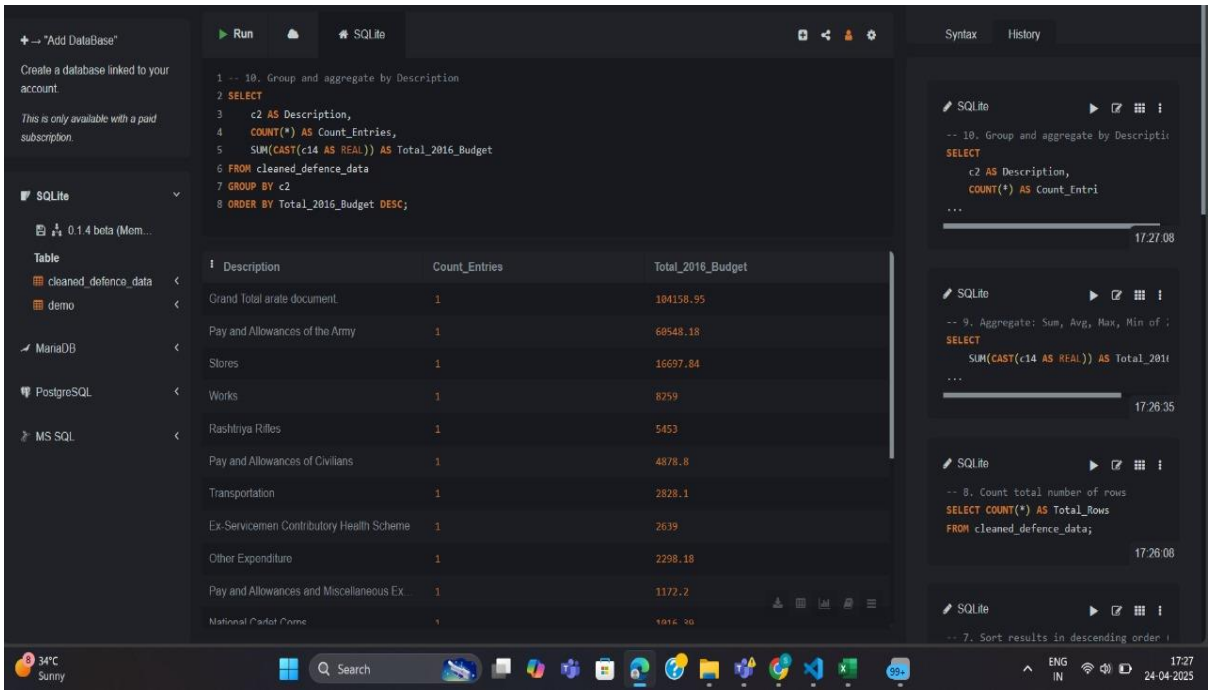
Table
cleaned_defence_data
demo
MariaDB
PostgreSQL
MS SQL
Run
SQLite
1 -- 9. Aggregate: Sum, Avg, Max, Min of 2016 Budget
2 SELECT
3 SUM(CAST(c14 AS REAL)) AS Total_2016,
4 AVG(CAST(c14 AS REAL)) AS Average_2016,
5 MAX(CAST(c14 AS REAL)) AS Max_2016,
6 MIN(CAST(c14 AS REAL)) AS Min_2016
7 FROM cleaned_defence_data;
Total_2016
Average_2016
Max_2016
Min_2016
208317.9
13887.859999999999
104158.95
-2836.88
Syntax
History
SQLite
-- 9. Aggregate: Sum, Avg, Max, Min of
SELECT
SUM(CAST(c14 AS REAL)) AS Total_201
...
17:26:35
SQLite
-- 8. Count total number of rows
SELECT COUNT(*) AS Total_Rows
FROM cleaned_defence_data;
17:26:08
SQLite
-- 7. Sort results in descending order
SELECT
c3 AS Scheme,
CAST(c6 AS
...
17:25:40
SQLite
-- 6. Combine conditions with AND/OR
34°C
Sunny
Search
ENG
IN
17:26
24-04-2025

10.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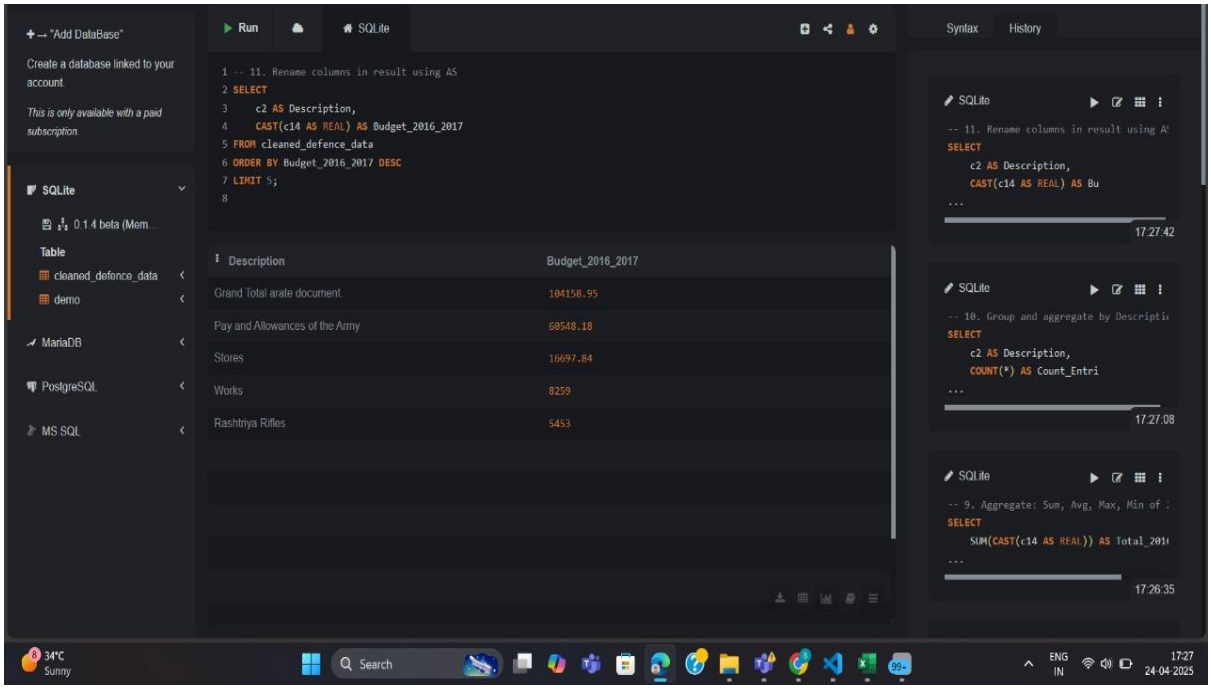
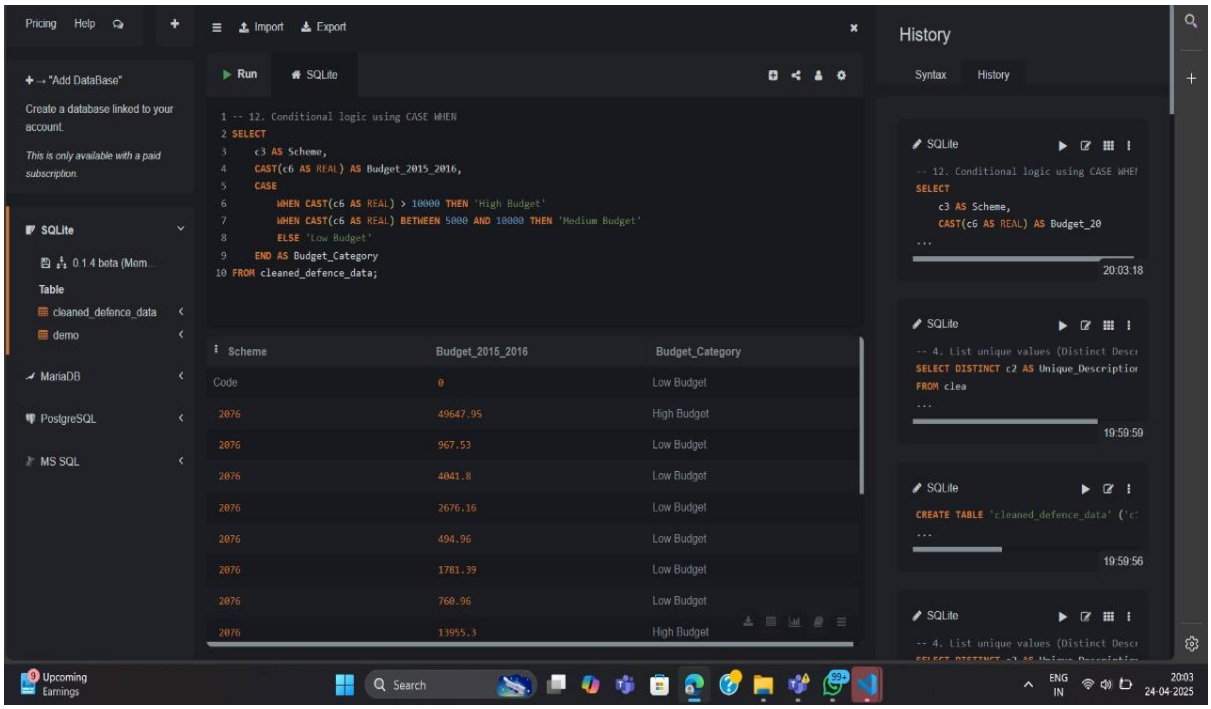
SQLite
0.1.4 beta (Mem...
Table
cleaned_defence_data
demo
MariaDB
PostgreSQL
MS SQL
Run
SQLite
1 -- 10. Group and aggregate by Description
2 SELECT
3 c2 AS Description,
4 COUNT(*) AS Count_Entries,
5 SUM(CAST(c14 AS REAL)) AS Total_2016_Budget
6 FROM cleaned_defence_data
7 GROUP BY c2
8 ORDER BY Total_2016_Budget DESC;
Description
Count_Entries
Total_2016_Budget
Grand Total arate document
1
104158.95
Pay and Allowances of the Army
1
60548.18
Stores
1
16697.84
Works
1
8259
Rashtriya Rifles
1
5453
Pay and Allowances of Civilians
1
4878.8
Transportation
1
2828.1
Ex-Servicemen Contributory Health Scheme
1
2639
Other Expenditure
1
2298.18
Pay and Allowances and Miscellaneous Ex...
1
1172.2
Syntax
History
17:27:08
17:26:35
17:26:08
34°C
Sunny
Search
ENG
IN
17:27
24-04-2025

11.


"Add DataBase"
Create a database linked to your account.
This is only available with a paid subscription
SQLite
0.1.4 beta (Mem...
Table
cleaned_defence_data
demo
MariaDB
PostgreSQL
MS SQL
Run
SQLite
1 -- 11. Rename columns in result using AS
2 SELECT
3 c2 AS Description,
4 CAST(c14 AS REAL) AS Budget_2016_2017
5 FROM cleaned_defence_data
6 ORDER BY Budget_2016_2017 DESC
7 LIMIT 5;
8
Description
Budget_2016_2017
Grand Total arate document
104158.95
Pay and Allowances of the Army
60548.18
Stores
16697.84
Works
8259
Rashtriya Rifles
5453
Syntax
History
17:27:42
17:27:08
17:26:35
34°C
Sunny
Search
ENG
IN
17:27
24-04-2025

12.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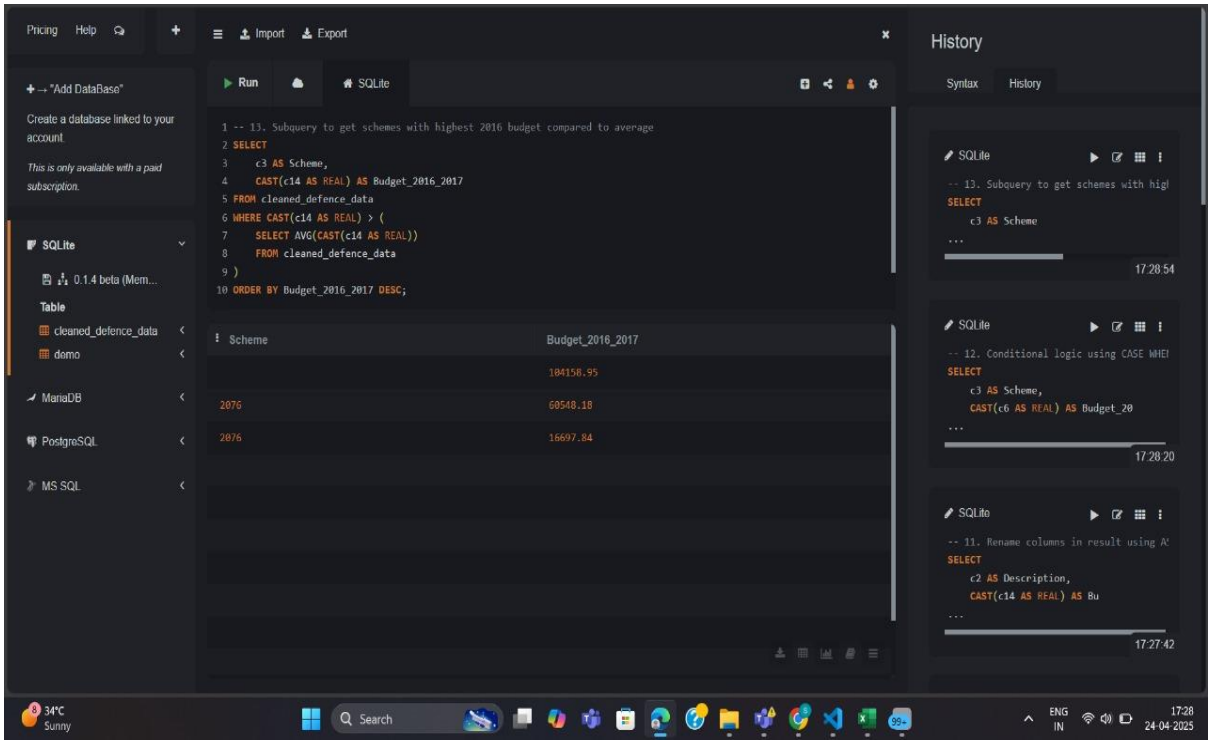
SQLite
0.1.4 beta (Mem...
Table
cleaned_defence_data
demo
MariaDB
PostgreSQL
MS SQL
-- 12. Conditional logic using CASE WHEN
SELECT
c3 AS Scheme,
CAST(c6 AS REAL) AS Budget_2015_2016,
CASE
WHEN CAST(c6 AS REAL) > 10000 THEN 'High Budget'
WHEN CAST(c6 AS REAL) BETWEEN 5000 AND 10000 THEN 'Medium Budget'
ELSE 'Low Budget'
END AS Budget_Category
FROM cleaned_defence_data;
Scheme
Budget_2015_2016
Budget_Category
Code
0
Low Budget
2076
49647.95
High Budget
2076
967.53
Low Budget
2076
4041.8
Low Budget
2076
2676.16
Low Budget
2076
494.96
Low Budget
2076
1781.39
Low Budget
2076
760.96
Low Budget
2076
13955.3
High Budget
History
Syntax
History
SQLite
-- 12. Conditional logic using CASE WHE
SELECT
c3 AS Scheme,
CAST(c6 AS REAL) AS Budget_20
...
20:03:18
SQLite
-- 4. List unique values (Distinct Descr
SELECT DISTINCT c2 AS Unique_Description
FROM clea
...
19:59:59
SQLite
CREATE TABLE 'cleaned_defence_data' ('c
...
19:59:56
SQLite
-- 4. List unique values (Distinct Descr
Upcoming
Earrings
Search
ENG
IN
20:03
24-04-2025

13.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SQLite
0.1.4 beta (Mem...
Table
cleaned_defence_data
demo
MariaDB
PostgreSQL
MS SQL
-- 13. Subquery to get schemes with highest 2016 budget compared to average
SELECT
c3 AS Scheme,
CAST(c14 AS REAL) AS Budget_2016_2017
FROM cleaned_defence_data
WHERE CAST(c14 AS REAL) > (
SELECT AVG(CAST(c14 AS REAL))
FROM cleaned_defence_data
)
ORDER BY Budget_2016_2017 DESC;
Scheme
Budget_2016_2017
104158.95
2076
60548.18
2076
16697.84
History
Syntax
History
SQLite
-- 13. Subquery to get schemes with hig
SELECT
c3 AS Scheme
...
17:28:54
SQLite
-- 12. Conditional logic using CASE WHE
SELECT
c3 AS Scheme,
CAST(c6 AS REAL) AS Budget_20
...
17:28:20
SQLite
-- 11. Rename columns in result using A
SELECT
c2 AS Description,
CAST(c14 AS REAL) AS Bu
...
17:27:42
34°C
Sunny
Search
ENG
IN
17:28
24-04-2025